Olivia Kaitlyn Roberts, MS – Curriculum Vitae

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EDUCATION

University of Michigan Aug 2024 – May 2026
Master of Science in Computational Epidemiology and Systems Modeling
Certificate in Complex Systems

University of South Dakota Aug 2021 – May 2024
Bachelor of Science in Mathematics, Summa Cum Laude
Minors in Music and Interdisciplinary Sciences

RESEARCH EXPERIENCE

Epidemiology Research Assistant, University of Michigan School of Public Health May 2025 – Present

- Conducts research under the mentorship of Dr. Andrew Brouwer at the Center for the Assessment of Tobacco Regulations (CAsToR)
- Applies multistate transition models using complex survey weights in R to analyze longitudinal trends in tobacco use among adults and youth in nationally representative studies
- Manages multiple research projects, conducts statistical analyses, and writes scientific manuscripts for publication
- Presents research findings at academic conferences

UDiscover Summer Research Scholar, University of South Dakota May 2023 – Aug 2023

- Conducted pure mathematics research with Dr. Gabriel Picioroaga on generalizing music theory using algebra

Behavioral Health Epidemiology Intern, South Dakota Department of Social Services May 2022 – Aug 2022

- Analyzed state-level data on substance use and mental health using R, Excel, and Power BI
- Contributed to behavioral health data and outcomes reports for state government and public use

TECHNICAL SKILLS

- Programming and Statistical Software: R, Python, Matlab, Stata, SQL
- Modeling: Markov multistate transition modeling, network modeling, deterministic and stochastic infectious and chronic disease modeling
- Tools: Microsoft Excel, Word, PowerPoint, Power BI, SEER Stat, LaTeX

PUBLICATIONS

- Brouwer AF, **Roberts OK**, Jeon J, Jimenez-Mendoza E, Land SR, Freedman ND, Torres-Alvarez R, Mistry R, Levy DT, Meza R. Patterns of cigarette and electronic nicotine delivery system transitions in U.S. adults, 2021–2024. *American Journal of Preventative Medicine*. 2026;108576. doi:10.1016/j.amepre.2026.108576.
- Picioroaga G, **Roberts O**. Musical systems with $\mathbb{Z}_n$ – Cayley graphs. *MusMat: Brazilian Journal of Music and Mathematics.* 2024;8(1). doi:10.46926/musmat.2024v8.1-23.

MANUSCRIPS IN PROGRESS

- **Roberts OK**, Jeon J, Jimenez-Mendoza E, Land SR, Freedman ND, Torres-Alvarez R, Mistry R, Levy DT, Meza R, Brouwer AF. Transitions in electronic nicotine delivery system and cigarette use among youth in the PATH Study from 2015–2024: a multistate transition modeling analysis. *Manuscript in preparation*

- **Roberts OK**, Bover Manderski MT, Gonsalves NJ, Jeon J, Meza R, Delnevo CD, Brouwer AF. Trends in transitions in cigarette and electronic nicotine vapor product use in the Rutgers Omnibus Study, 2022–2026. *Manuscript in preparation*
- **Roberts OK**, Sanchez-Romero LM, Jimenez-Mendoza E, Jeon J, Travis N, Mistry R, Meza R, Levy DT, Brouwer AF. Rates of oral nicotine pouch experimentation, initiation, and cessation by other tobacco product use and sociodemographic characteristics among U.S. adults, 2022–2024. *Manuscript in preparation*

PRESENTATIONS

"Trends in transitions in cigarette and electronic nicotine vapor product use in the Rutgers Omnibus study, 2022–2025"

- Poster presentation, University of Michigan Epidemiology Poster Session, Apr 2026
- Poster presentation, NIH Tobacco Regulatory Science Meeting, July 2026

"Transitions in cigarette, ENDS, and dual use among youth in the PATH Study from 2015–2023: a multistate transition modeling analysis"

- Invited seminar talk (1 hour), University of South Dakota Department of Mathematics, Mar 2026
- Oral presentation, Society for Research on Nicotine and Tobacco (SRNT) Meeting, Mar 2026

"Musical Systems with $\mathbb{Z}_n$ – Cayley Graphs"

- Oral presentation, Joint Mathematics Meetings (JMM) Pi Mu Epsilon Session, Jan 2024
- Oral presentation, Nebraska Conference for Undergraduate Women in Mathematics (NCUWM), Jan 2024
- Oral presentation, National Conference on Undergraduate Research (NCUR), Apr 2023 and Apr 2024

HONORS & AWARDS

Research Awards and Grants

- Pi Mu Epsilon Outstanding Student Speaker Award, Joint Mathematics Meetings, Jan 2024
- Undergraduate Research Excellence Award, University of South Dakota, Sept 2023
- UDiscover Summer Research Grant, University of South Dakota, May 2023
- John W. Carlson Research Grant, University of South Dakota, Apr 2023

Scholarships

- Epidemiology Student Scholarship, University of Michigan, Feb 2024 (75% of tuition)
- South Dakota NASA Space Grant Consortium Educational Stipend, Oct 2023
- Dr. Cleland Cook Mathematics Scholarship, University of South Dakota, Apr 2022 and Apr 2023

TEACHING EXPERIENCE

Yoga Teacher, University of Michigan Recreation — Aug 2024 – Present

- Leads restorative and vinyasa yoga classes to 20+ students multiple times a week
- Developed a 6-week Elevate Yoga workshop series to deepen students' yoga practice

Instructor Aide, University of Michigan Department of Mathematics — July 2024 and July 2025

- Co-instructed three summer camp sessions for 15-20 high school students for the Michigan Math and Science Scholars program
- Assisted with curriculum development and delivered lectures on group theory, set theory, and music theory

Instructor Assistant, University of Michigan School of Nursing — Jan 2025 – May 2025

- Supported 100 students in NURS 218 (Applied Statistics)
- Held office hours, graded assignments, assisted students with Stata programming, and led review sessions

Math Tutor, University of South Dakota Jan 2022 – May 2024

- Tutored undergraduate students in algebra, statistics, and calculus

Yoga Teacher, University of South Dakota Wellness Center Aug 2021 – May 2024

- Led sculpt and vinyasa yoga classes for 20+ students multiple times per week
- Organized the first and second annual USD Yoga Festivals and led multiple yoga workshops

OTHER

Event Volunteer, VegMichigan June 2024 – Present

CPR/AED for Professional Rescuers Certification Aug 2026

U-M Summer Session, Simulation Modeling of Tobacco Use, Health Effects and Policy Impacts July 2026

Admissions Ambassador Volunteer, University of Michigan School of Public Health Aug 2024 – May 2026

200-Hour Yoga Teaching Certification, Tiny Buddha Yoga, Ann Arbor, MI Dec 2024

200-Hour Yoga Teaching Certification, Better Together Yoga, Sioux Falls, SD Sept 2021